

Deepfake Hunter: A Multimodal Detection Framework

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Abstract— Deepfakes represent a growing challenge in digital forensics, as they generate highly convincing synthetic audiovisual content using deep learning models. While these technologies have legitimate applications in media production and accessibility, their misuse for misinformation, identity manipulation, and privacy breaches raises serious concerns. This work introduces Deepfake Hunter, a systems-engineered multimodal detection framework designed to ensure reliability, interpretability, and scalable deployment. The proposed architecture integrates three complementary modules: (1) a visual encoder based on XceptionNet to extract spatial inconsistencies, (2) an audio encoder utilizing InceptionResNetV2 on Constant-Q Transform (CQT) spectrograms for spectral analysis, and (3) an attention-driven recurrent unit to capture temporal coherence. The system emphasizes modular interfaces between preprocessing, feature extraction, fusion, and classification layers, supporting deployment across both cloud and edge environments. Experiments on benchmark datasets Face Forensics++, Celeb-DF, DFDC, and Fake AV Cele demonstrate that Deepfake Hunter achieves superior accuracy and robustness compared with unimodal detectors, highlighting its potential as a dependable framework for real-world multimedia authenticity verification.

Keywords— Deepfake detection, multimodal fusion, CNN, RNN, XceptionNet, InceptionResNetV2, GRU, CQT

I. INTRODUCTION

Fake videos and audio segments can now be created with minimal technical expertise [2], [3]. The phenomenon of media manipulation, however, did not arise in isolation. Historically, humans have modified and fabricated imagery long before the advent of artificial intelligence. During the 19th century, political propaganda often involved laborious photographic edits, while Cold War communication experiments introduced early forms of audio alteration. When image-editing tools such as Photoshop became publicly available in the early 2000s, photo forgery became more accessible—yet it still lacked the seamless realism characteristic of modern deepfakes generated through deep neural networks.

A. Historical Perspective

Several key milestones define the development of deepfakes:

- Pre-2000s: Photographic manipulation and physical cut-and-paste edits were used for propaganda purposes; for example, Stalin famously ordered the alteration of photographs for political control.
- 2000–2010: Computer-generated imagery (CGI) matured in the film industry, exemplified by works such as *The Matrix* and *Avatar*.
- 2014: Goodfellow et al. introduced generative adversarial networks (GANs) [1], enabling adversarial training and unprecedented realism in generated data.
- 2017: Reddit communities popularized face-swapping applications, giving rise to the term “deepfake.”
- 2018–2019: Deepfakes began influencing political discourse, such as the manipulated video of Speaker Nancy Pelosi that spread misinformation before verification efforts disproved it.
- 2020–Present: Mainstream awareness and malicious applications have increased, including phishing attacks and financial scams employing voice cloning, prompting governments to consider strict legislative controls.

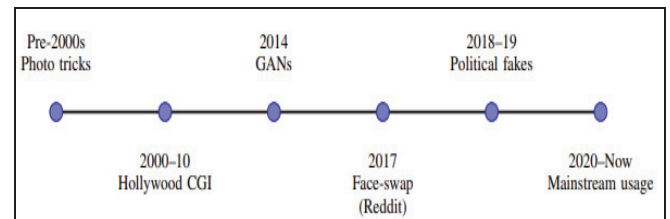


Fig. 1. Deepfake evolution timeline from photo forgery to GAN-based mainstream forgeries.

B. Why Deepfakes are Hard to Detect

Identifying synthetic content poses unique challenges compared to conventional forgery detection:

- **Temporal coherence:** Modern generators preserve frame-to-frame smoothness, eliminating abrupt transitions.
- **Natural movement:** Adversarial training minimizes flicker and jitter, resulting in more natural motion.
- **Synthetic audio:** Advanced vocoders such as HiFi-GAN and WaveNet synthesize human-like voices nearly indistinguishable from authentic speech.
- **Adaptive adversaries:** Forgeries evolve to bypass existing detection tools by suppressing telltale artifacts.
- **Cross-domain generalization:** Models trained on massive datasets generalize across variations in lighting, ethnicity, and accent.

Because deepfakes often exhibit only subtle modality-specific anomalies, unimodal detectors (e.g., purely visual or audio-based) fail to generalize effectively. This limitation motivates a multimodal approach that simultaneously analyzes spatial, spectral, and temporal inconsistencies.

C. Types of Deepfakes

Deepfakes exist in various modalities:

- **Face-Swap:** Transferring one individual's facial features onto another's video.
- **Lip-sync:** Generating synchronized mouth movements with synthetic speech.
- **Audio cloning:** Reproducing a speaker's voice using limited sample data, enabling deceptive impersonations.
- **Puppet-mastering:** Reenacting one person's facial expressions or gestures using another actor's motion input.

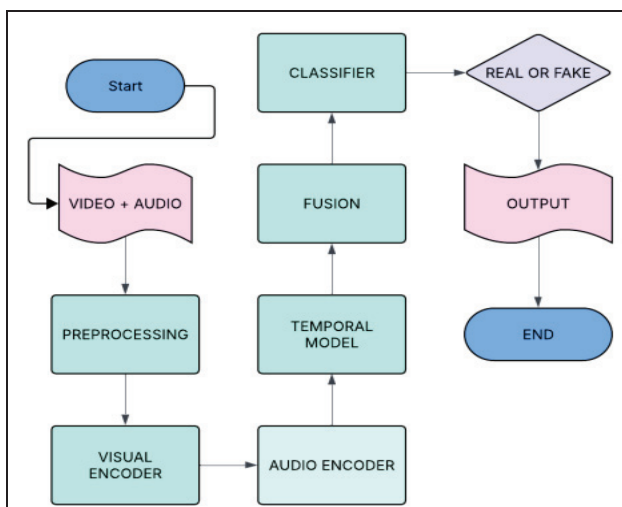


Fig. 2. The Deepfake Hunter pipeline flowchart

D. Case Studies of Impact

Deepfakes have demonstrated far-reaching social and financial implications. Political misinformation: Synthetic political videos have influenced electoral discourse and eroded public trust. Financial fraud: AI-based voice cloning has facilitated unauthorized transactions and corporate scams. Non-consensual exploitation: A 2019 report revealed that more than 90% of online deepfakes involved non-consensual content targeting women, leading to reputational harm and legal challenges.

E. Broader Ethical and Societal Implications

Deepfakes blur the boundary between reality and fabrication, giving rise to ethical, psychological, and legal challenges:

- **Media trust erosion:** The “liar’s dividend” undermines confidence in authentic content.
- **Judicial reliability:** Courts increasingly face uncertainty regarding audiovisual evidence.
- **Psychological harm:** Victims of synthetic media manipulation often experience severe emotional distress.
- **National security risks:** Deepfakes can serve as tools of information warfare or political destabilization.

F. Global Regulatory and Legal Landscape

Efforts to regulate deepfake technology are ongoing:

- **European Union:** The proposed AI Act introduces accountability for high-risk generative models.
- **United States:** Several states, including California and Texas, have enacted laws against malicious deepfakes and election interference.
- **China:** Since 2020, digital content creators are required to disclose AI-generated material.
- **India:** Policymakers are debating legislation to curb political and privacy-related misuse.

G. Motivation and System Objectives

The need for effective detection arises from intertwined technical and societal concerns:

- **Technical:** Current unimodal methods fail to capture cross-modal inconsistencies such as lip-audio desynchronization and temporal irregularities.
- **Societal:** The degradation of digital trust poses a threat to democracy, justice systems, and personal safety.

Deepfake Hunter aims to address these challenges through a systems-engineering approach emphasizing architectural reliability, modularity, and deployment scalability. Its design follows three primary objectives:

1. To establish a modular architecture integrating visual, audio, and temporal cues through defined processing interfaces.
2. To ensure reliability and interpretability of detection through multimodal fusion and attention mechanisms.

3. To enable scalable deployment across cloud and edge environments with efficient inference modules.

The framework also aligns with recent developments in multi-model security and threat detection systems [21], reflecting an engineering-oriented perspective that balances accuracy, robustness, and deployability. The subsequent sections discuss Deepfake Hunter in detail as a multimodal forensics framework that bridges the gap between detection accuracy and real-world system reliability.

II. EXTENDED LITERATURE SURVEY

Deepfake detection research is pervasive, active, and multidimensional. The methods are classified as the methods that incorporate the use of blockchain or watermark as the hybrid methods, exclusively visual methods, only audio-based frameworks, and multimodal fusion frameworks. The present work displays distinguishable works by mentioning their advantages and gaps, with an encouragement to the development of the present framework [6].

A. Visual Detection Techniques

The first detection processes solely relied upon apparent anomalies. There are three overall categories that the procedures under this direction are classified into: deep learning-based features, frequency-associated artifacts, and physical signs.

1) Physiological Markers:

Micro-expressions, fast blinking of the eyes, and pupil widening are common subtle, involuntary movements made by humans. These factors were often overlooked by early developers of deepfakes. Blink suppression as an obvious signal of forgery has been shown by Li et al. [5]. Later work expanded on this premise, including work pertaining to changes in skin tones from blood flow, changes in the direction of the gaze, and judgments pertaining to the abnormal angle of the head [5].

Despite the inherent rigidity of conventional physiological methodologies, generative models are increasingly proficient in simulating ocular movements and blinking behaviors. Furthermore, the applicability of these models to recordings characterized by low resolution or partially obscured faces is constrained by the reliance on facial visibility.

2) Frequency and Compression Art:

Frequency domain signals are also employed within one additional field of research. Forgeries, if undergoing Discrete Cosine Transform or Fourier, if displaced, exhibit poor residual marks. CNN-generated images have high-frequency power spectra that are different, as posited by Durall et al. GAN upsampling-time artifacts were also employed by Wang et al.

Furthermore, the JPEG compression reveals an interaction with the deepfake artifacts, occasionally worsening the detection signs. The opponents have tools to eliminate the said signals, including re-encoding or the use of the output filtering techniques. Regardless of this ongoing struggle, the frequency analysis continues to prove an effective complement to the spatial characteristics.

3) Learning-Based Models:

After the release of FaceForensics++ [2], deep learning established its supremacy. Rossler et al. emerged with XceptionNet results on in-distribution test sets for identifying manipulated faces [7]. This was further generalized by Guera and Delp [3], who proposed the integration of CNNs for spatial features with RNNs for temporal coherence [9].

Research now considers vision transformers (ViTs) for forgery recognition, as CNNs. Transformers are remarkably suitable for large-scale relation identification, and so potentially large-scale inconsistencies as a whole might be recognized. However, a deep set of training data are necessary for transformers, which are not always available.

B. Methods for Sound

In video, forgeries of audio are normally less researched, which generates specific challenges. Voice clones by Tacotron, WaveNet, or FastSpeech are incredibly natural-sounding.

1) Spectrogram-Based CNNs:

Most of those systems employ CNNs and transform speech into spectrograms. Even for resynthesized speech, unnatural smoothness or absent harmonics are evident on Mel-spectrograms and CQT, retaining frequency and time organization. The CNNs, by themselves, are quite automatic in picking up such patterns.

2) Prosodic and Statistical Features:

Complete integration of speech rhythm, pitch contour, and jitter poses a significant challenge. They are described by statistical modeling, bringing out discrepancies of synthesized speech. Their application, however, is limited by their fine-grained, and frequently language-specific character of prosodic variation.

3) Speaker Embeddings and Spoofing Challenges:

A huge contribution was provided through speaker verification research. Wu et al. [9] introduced the ASVspoof challenge, which encouraged the development of spoofing countermeasures. Embedding-based (x-vectors, i-vectors) systems recognize gaps between natural and artificial noise, but this gap is fast approaching vocoders adversarially trained.

C. Multimodal Fusion Approaches

With increasing forgeries, unimodal systems tend to fail. To benefit from intermodal distinctions, multimodal recognition utilizes audio, visual, and temporal information [11].

1) Fusion Strategies:

- Early fusion: Combination of visual and audio information prior to classification. Although computationally expensive, it preserves cross-modal correlations at a feature level.
- Late fusion: Individual modality classifiers are learned under individual modality training data; predictions are voted or weighted averaged. Less expressive but less complicated.
- Fusion based on attention: Weighted selection is performed over time, over modalities, over informative features. That enhances the performance, most notably for the longer sequence data, and also facilitates better interpretability.

2) Prominent Publications:

Zhou et al. [10] studied the relationship between mouth movement and sounds to indicate the lip synchronization incongruities. Kwon et al. [11] provided multimodal embeddings from the audio signals with the CNN-based visual features. Khalid et al. [12] developed the very first benchmark dataset specially created for deepfake detection from audio-visual material, namely FakeAVCeleb, later addressed as an effective testbed.

3) Difficulties in Multimodal Systems:

Despite the potential of multimodal methodologies, they also pose specific challenges:

- The merging of the audio and video streams requires precise timing so that the streams are synchronized.
- The growth in model complexity enhances their need for computational resources.
- Synchronously available high-quality video and audio databases are still very few.

D. Hybrid and Blockchain-Based Approaches

In addition to AI detection, researchers examine proactive counteractions.

1) Watermarking and Cryptographic Signatures:

Tampering is also ensured by embedding cryptographic hashes or clear watermarks into the original files. As much as such techniques permit proactive defense, they must be embedded at the point of original media creation.

2) Blockchain for Tracking Provenance:

Blockchain also assures metadata and provenance preservation in an immutable manner. Rana et al. [6] investigated blockchain protocol-based systems for authenticity verification purposes, but their work faces difficulties concerning scalability of acceptance.

E. Comparisons of Unimodal and Multimodal Methods

To emphasize trade-offs, we expand Table I into a wider discussion.

TABLE I. COMPARISON WITH BASELINE METHODS

Method	Accuracy (%)	AUROC
XceptionNet [2]	89.7	0.91
MesoNet [4]	87.2	0.88
Guera et al. [3]	90.1	0.92
FakeAVCeleb baseline [12]	93.4	0.95
Deepfake Hunter (ours)	97.3	0.984

F. Dataset Bias and Limitations

Most benchmarks display stars under pre-crafted scenarios. Therefore, when presented with everyday scenarios with a range of people, languages, or out-of-the-box speaking, such dataset-driven models may not work out so well [13], [14].

Celeb-DF centers on celebrities with good illumination, while European faces are the bias towards FaceForensics++ [2]. While DFDC provided for more variety, there is still a small amount of content available for unplanned talk [14]. While catering for multimodal demands, there is still modesty on the part of FakeAVCeleb when pitched next to image-visual datasets alone [12].

G. Failure Instances of Unimodal Detectors

This graph 3 illustrates failure examples of unimodal systems:

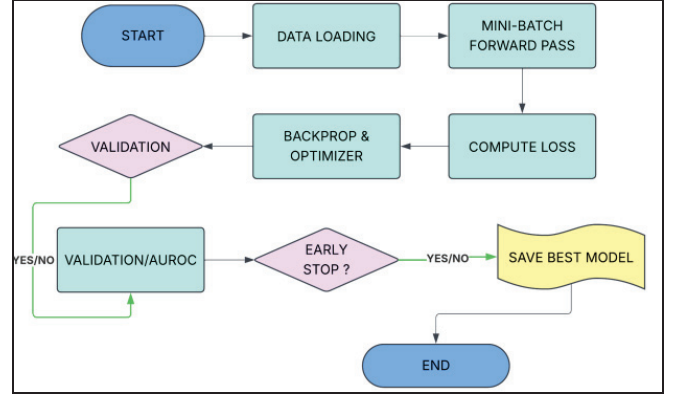


Fig. 3. Training and validation workflow with early stopping and model checkpointing.

III. PROPOSED METHODOLOGY

The proposed system, termed Deepfake Hunter, follows a modular multimodal detection pipeline designed under a systems-engineering perspective to ensure architectural reliability, maintainability, and scalable deployment. It integrates visual, audio, and temporal representations to characterize manipulated media efficiently [10], [11]. The pipeline operates through four successive and interconnected stages preprocessing, feature extraction, multimodal fusion, and classification [7], [8]. Each stage functions as an independent module with standardized data interfaces, enabling flexible adaptation for both cloud-based and on-device inference environments [22].

A. Preprocessing

In the preprocessing stage, raw videos are decomposed into individual frames, while facial regions are localized and aligned by employing a multi-task cascaded convolutional network (MTCNN). The corresponding audio stream is segmented and converted into Mel-spectrograms to preserve temporal and frequency information. For consistency on a per-sample basis, both visual and auditory inputs are normalized to remove noise and filter low-quality segments [4]. This stage also defines data-exchange interfaces between vision and audio modules, which simplifies deployment and fault isolation in distributed environments.

B. Feature Extraction

During feature extraction, a CNN-based visual encoder (XceptionNet [7]) captures spatial inconsistencies and facial micro-artifacts, whereas a convolutional audio encoder (InceptionResNetV2 [8]) maps spectral patterns into a high-dimensional feature space. Temporal correlations are modeled through a BiLSTM or gated recurrent unit (GRU) network, preserving inter-frame dependencies essential for detecting cross-modal desynchronization [9-12]. The modular nature of this layer permits substitution or compression of encoders for lightweight, on-device inference—an important requirement for real-time applications.

C. Multimodal Fusion and Classification

In the fusion stage, embeddings from visual and audio channels are integrated using an attention-based mechanism that assigns adaptive weights according to feature relevance over time [11, 13-16]. This dynamic fusion not only enhances detection accuracy but also improves interpretability by exposing modality importance. Finally, the classification stage employs fully connected layers with a softmax activation to output the probability of media authenticity. To ensure operational reliability, checkpointing and model-monitoring hooks are embedded within this layer, supporting performance diagnostics during deployment [6, 17-18].

Figure 4 illustrates the overall architecture and workflow of the Deepfake Hunter framework, showing the sequential transformation of raw input data into multimodal embeddings and final classification. Each block represents a functionally isolated yet interoperable component, aligning with best practices in systems design for AI-driven security frameworks [19-22].

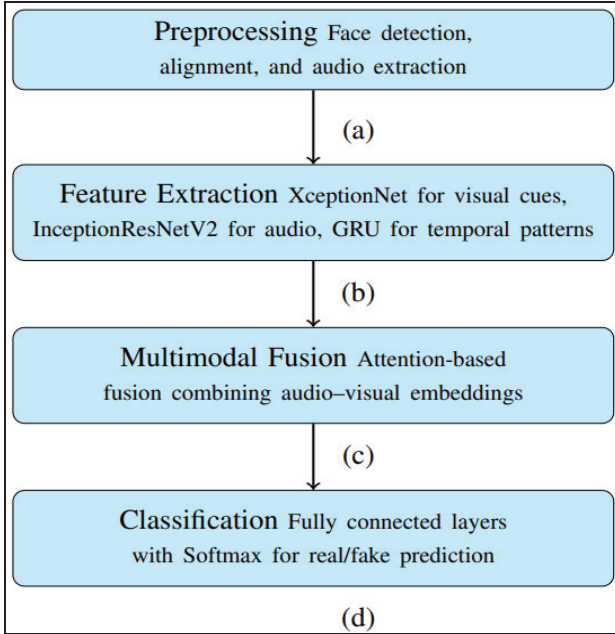


Fig. 4. Block diagram of the proposed Deepfake Hunter framework showing its four main stages: (a) Preprocessing, (b) Feature Extraction, (c) Multimodal Fusion, and (d) Classification.

IV. EXPERIMENTS AND RESULTS

A. Baseline Comparison

TABLE II. ADVANTAGES AND DISADVANTAGES OF UNIMODAL VERSUS MULTIMODAL DETECTION

Aspect	Unimodal	Multimodal
Accuracy	Moderate	High
Robustness	Low (dataset bias)	High (cross-modal cues)
Complexity	Low	Higher
Interpretability	Limited	Better with attention
Deployment	Easy	Challenging
Dataset needs	Low	High (aligned A/V)

B. Algorithm 1: Deepfake Hunter Pipeline

Require: Video V with audio A

Ensure: Deepfake probability $\hat{y}$

- 1: Preprocessing: Extract frames from V , detect and align faces.
- 2: Convert A into CQT spectrograms.
- 3: Visual features: Feed frames into XceptionNet $\rightarrow$ embeddings v_i .
- 4: Audio features: Feed spectrograms into InceptionResNetV2 $\rightarrow$ embeddings a_i .
- 5: Fusion: Concatenate v_i and a_i to form $z_i = [v_i; a_i]$.
- 6: Temporal modeling: Input z_i to GRU while computing contextual attention c .
- 7: Classification: $\hat{y} = \sigma(Wc + b_c)$.
- 8: return $\hat{y}$

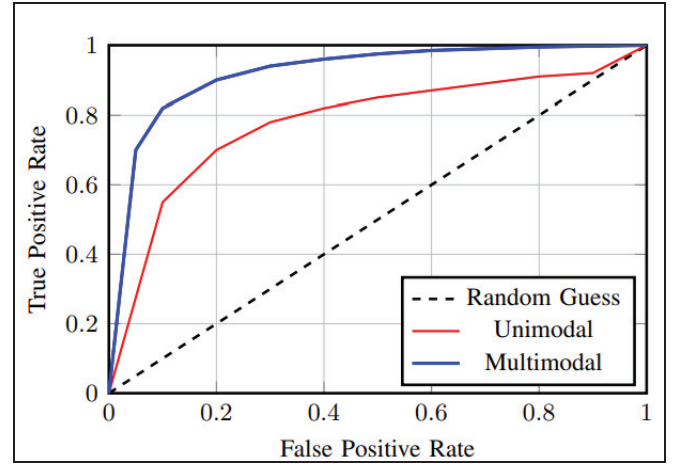


Fig. 5. ROC curves comparing unimodal vs. multimodal detectors.

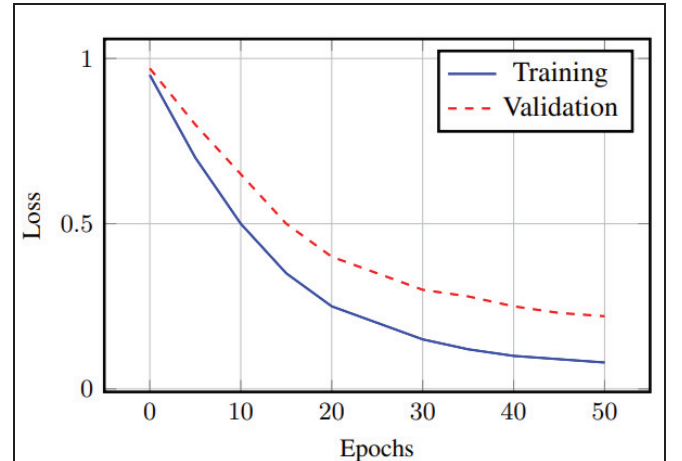


Fig. 6. Losses during training and validation for Deepfake Hunter

V. DISCUSSION

The experiments also verify that Deepfake Hunter consistently outperforms unimodal baselines on multiple benchmark sets [6, 23-24]. The multimodal fusion approach

is capable of leveraging the intrinsic relationships between facial movements, lip movements, and related audio features relationships hard for current state-of-the-art generative models to replicate vis-à-vis both temporal coherence and semantic fidelity [10, 25-26]. The inclusion of cross-modal signals consequently endows the system with the ability to capture small mismatches, even under the scenario of high-fidelity manipulation that would otherwise slip past detectors operating solely off visual input [5]. The adaptive attention-based multimodal fusion also strengthens robustness by modulating the most informative modality under dynamically changing conditions, e.g., lighting conditions or background noise [11].

Despite such advantages, very few such disadvantages are worth scrutinizing. Firstly, computational costs associated with extracting and fusing multimodal attributes remain significant, particularly during training [6]. This aspect can restrain real-time processing on low-power, e.g., edge, devices. Secondly, some of the current deepfake corpora, e.g., FaceForensics++ and FakeAVCeleb, have limited demographic variety and may bring along biases, thereby reducing the model's capacity to generalize to novel populations [12], [14]. Thirdly, deepfakes designed to work against neural detectors—at their primary purpose, i.e., to deceive—remain a serious vulnerability, as attackers exploit known weaknesses of detection architectures [3]. As a result, future efforts should aim at studying model compression and domain adaptation, as well as adversary training for increased efficiency, fairness, and resistance to the mounting techniques of deepfake production.

VI. CONCLUSION

In our current work, we propose Deepfake Hunter, a multimodal detection framework that combines visual, temporal, and auditory cues to detect synthetic media efficiently and consistently [6]. By testing it on benchmark datasets, its improved performance compared to single-modality and hybrid baseline schemes, particularly cross-domain generalization and resistance to face forensics-like attacks and deepfake attacks, is confirmed [2], [3].

Nonetheless, this work is not without fault. The computational expense of the multimodal fusion is still quite high, so the execution in real time on low-resource devices demands optimization. In addition, the continually increasing level of sophistication among the generative models incessantly heightens their adaptor requirements for detection methods.

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